

Monocular Depth Estimation: A Thorough Review

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(Survey Paper)

Abstract—Estimation of depth in two-dimensional images is among the challenging topics in Computer Vision. This is a well-studied but also an ill-posed problem, which has long been the focus of intense research. This paper is an in-depth review of the topic, presenting two aspects, one that considers the mechanisms of human depth perception, and another that includes the various Deep Learning approaches. The methods are presented in a compact and structured way that outlines the topic and categorizes the approaches according to the line of research followed in the recent decade. Although there has been significant advancement in the topic, it was without any connection with human depth perception and the potential benefits from this sector.

Index Terms—Artificial intelligence, computer vision, deep learning, depth estimation, image processing, machine learning, monocular depth.

I. INTRODUCTION

RECOGNITION of the spatial structure of a scene is a fundamental problem in machine visual perception, typically called *depth estimation* or *reconstruction*. When a 3-dimensional (3D) scene is projected onto a plane, a 2-dimensional (2D) image arises. The goal of depth estimation is to solve the inverse problem, which is to reconstruct the lost dimension that corresponds to the distance between the scene objects and the viewing location. The methods that have been proposed thus far for the representation of the spatial structure and the mapping between a 3D scene and its corresponding 2D image belong to the field of depth estimation [1].

Formally, given a single image $I \in \mathbb{R}^{M \times N \times 3}$ with three chromatic channels and size $M \times N$, the goal is to estimate a depth map $D \in \mathbb{R}^{M \times N}$. Dealing with the issue as an image reconstruction problem, a function $f: I \rightarrow D$ is needed, to map the scene's depth.

Research in this topic has an immediate impact on a wide spectrum of applications, such as image segmentation [2], autonomous navigation [3], further enhanced with object detection

and obstacle avoidance [4], robot-assisted surgery [5], detection of robotic grasps [6], human pose estimation [7], automatic film conversion from 2D to 3D [8], synthetically defocusing of images [9], and synthetic object insertion into images [10]. Although these applications had a tremendous improvement in recent years due to the advancements in machine learning and artificial intelligence, the attained visual perception is, yet, far from beating the human skill.

Our current understanding of *human depth perception* is that it lies within the exploitation of information drawn from visual cues [11], [12], [13]. A visual cue for depth can be thought of as differential information that arises from the complex human visual perception that constantly compares visual input from the eyes.

Despite humans' inherent ability to perceive spatial structure, most technical approaches still find it very challenging to acquire general depth perception. Today, several methods for depth estimation exist, which rely on a limited number of depth cues like stereo [14], monocular (e.g., texture variations, texture gradients, haze, relative sizes of objects and occlusions) [15], motion parallax [16], and focus [17]. Furthermore, there is a fundamental distinction between the methods as to whether they are capable of understanding an absolute measurement or a relative depth estimate of the observed scene. In some applications, the relative depth estimate may be sufficient, while in others the absolute depth measurement is deemed necessary. Thus, in most cases, the application domain determines the type of method to be used. Recently, the availability of computing resources, power, and memory, as well as the advancement of Deep Learning techniques, became encouraging factors in revisiting the depth estimation problem.

In the field of Deep Learning-based monocular depth estimation, several reviews have been published, highlighting the new methods, the challenges and future directions. Xiaogang et al. [18] conducted a review that focused on performance evaluation and future directions. They compared a small number of methods based on evaluation metrics and discussed the potential of adaptive methods, high-density image depth estimation using efficient network architectures, and the challenge of accurate depth estimation in complex scenes. Masoumian et al. [19] discussed Convolutional Neural Network (CNN) architectures and the need of advancements in monocular depth estimation. They highlighted the general problem of limited datasets with groundtruth, and proposed six objectives for future research: accuracy, computational time, resolution quality, real-time inference, transferability, and input data shapes. Ming et al. [20]

Manuscript received 8 February 2023; revised 16 October 2023; accepted 4 November 2023. Date of publication 8 November 2023; date of current version 6 March 2024. Recommended for acceptance by P. Mordohai. (Corresponding author: Vasileios Arampatzakis.)

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Digital Object Identifier 10.1109/TPAMI.2023.3330944

identified common features of current methods, which are the multiple layers and the inherent problem of information loss in encoder-decoder architectures. They outlined five main challenges for future methods, including the integration and optimization of network frameworks, dataset construction, handling dynamic objects and occlusions, generating high-resolution depth maps, and achieving real-time performance. Poggi et al. [21] focused on the binocular methods for stereo-based depth estimation. They summarized recent achievements and emphasized two future directions: generalization across different domains and the applicability of methods to high-resolution images. Zhao et al. [22] discussed the future development of monocular depth estimation along three main axes: accuracy, transferability, and real-time performance. They also highlighted the lack of research on the underlying mechanisms of Deep Learning-based monocular depth estimation methods, such as understanding what depth cues they exploit and what the networks have learned.

The motivation for this work was that, despite the rapid advancements in monocular depth estimation, progress seems to follow a trial and error pattern, without any attention on explainability and connection with human depth perception mechanisms. To this end, we review, categorize and discuss the relevant research, providing a twofold contribution: (a) to describe the problem in detail and catalog the current state-of-the-art, and (b) to highlight the need for further study in terms of understanding what a method is able to provide and how (explainability [23]). The most typical methods used in enabling explainability in various AI domains include (a) feature visualization [24], (b) saliency maps [25], (c) attention mechanisms [26], (d) LIME (Local Interpretable Model-agnostic Explanations) [27], (e) SHAP (Shapley Additive exPlanations) [28], and (f) Grad-CAM (Gradient-weighted Class Activation Mapping) [29]. In addition, one has to pay attention to trade-offs between simplicity and accuracy in explanation methods, the fact that some methods are model-specific, while others are model-agnostic, and that there is a need to validate explanations and ensure they reflect true model behavior. Limited research has been conducted to this end in monocular depth estimation, as discussed in section IV. In this study, we propose to also define explainability (in this domain) as the *understanding of the connection between the depth predictions by AI systems and the human depth estimation mechanism*, and suggest that future research on this direction may be beneficial.

The paper starts by presenting the human visual depth cues, and then moves to the study of conventional and advanced methods for depth estimation, summarizing the relevant benchmark datasets and the applicable evaluation metrics. The paper concludes with remarks highlighting future perspectives in tackling monocular depth estimation.

II. BACKGROUND AND RELATED WORK

Depth perception by mechanical systems has long been a major challenge for the scientific community. In his treatise “Optics”, Euclid was among the first to produce theorems for this topic; “objects of equal size unequally distant appear unequal and the one lying nearer to the eye always appears larger”, and

“when objects move at equal speed, those more remote seem to move more slowly” are just two of those theorems [30]. Without a doubt, Euclid implies that the *relative size* of objects and the *motion parallax* are two significant visual depth cues, in human depth perception, as will be described in the following texts. Since then, and particularly for the *monocular depth perception*, various depth cues have been proposed, such as the perspective, the relative scale, the appearance under controlled lighting and shading, the form of the shadows, and the presence of occlusions [31]. This review follows the research of Cutting & Vishton [11], as described in Section II-A.

A. Human Depth Perception

It was as early as 1995 when Cutting & Vishton [11] published their detailed study and conclusions on human depth perception. They described and categorized all the visual depth cues that Nagata [12] had previously proposed. They identified nine primary depth cues, which are graphically depicted in Fig. 1, and defined as follows:

- 1) *Relative Size* (RS) refers to the apparent difference in the area occupied by the projection of objects of physically similar size on the retina. In other words, between two similar objects, the one that seems smaller is farther from the observer.
- 2) *Relative Density* (RD) is detected when there is a cluster of similar objects or textures. As the distance from the observer increases, the objects appear to be closer to each other and the textures become denser.
- 3) *Height in the visual field* (H) is connected with the relation of the bases of objects, assuming the presence of a ground plane and the absence of a ceiling [32]. Specifically, objects at a greater distance from the observer tend to appear closer to the horizon line.
- 4) *Aerial Perspective* (AP) appears when the air has a high concentration of moisture, pollution, or both, and the objects in the distance become bluer, have less contrast, or both in comparison to those in the foreground.
- 5) *Motion Perspective* (MP) is the phenomenon that appears when there is relative motion between the observer and observed objects. Objects that appear to be moving faster are closer to the observer.
- 6) *Convergence* (C) is connected with the angle between the optical axes of the two eyes. If an object is far from the observer, the optical axes tend to be parallel and the angle between them is close to zero degrees. As the object distance decreases, the two axes converge to a point on the object.
- 7) *Accommodation* (A) is the eye function in which the lens (which is essentially an asymmetric biconvex lens) changes its shape to enable focusing at different distances.
- 8) *Binocular Disparity* (BD), the bases for what is called stereo vision, is the difference in the location of an observed object in the two eyes and only has meaning if more than one viewing sensors (eyes in this case) are available.

The combination of the visual depth cues seems to be associated with full scene understanding by the humans. Furthermore, Cutting & Vishton [11] mentioned secondary cues, such

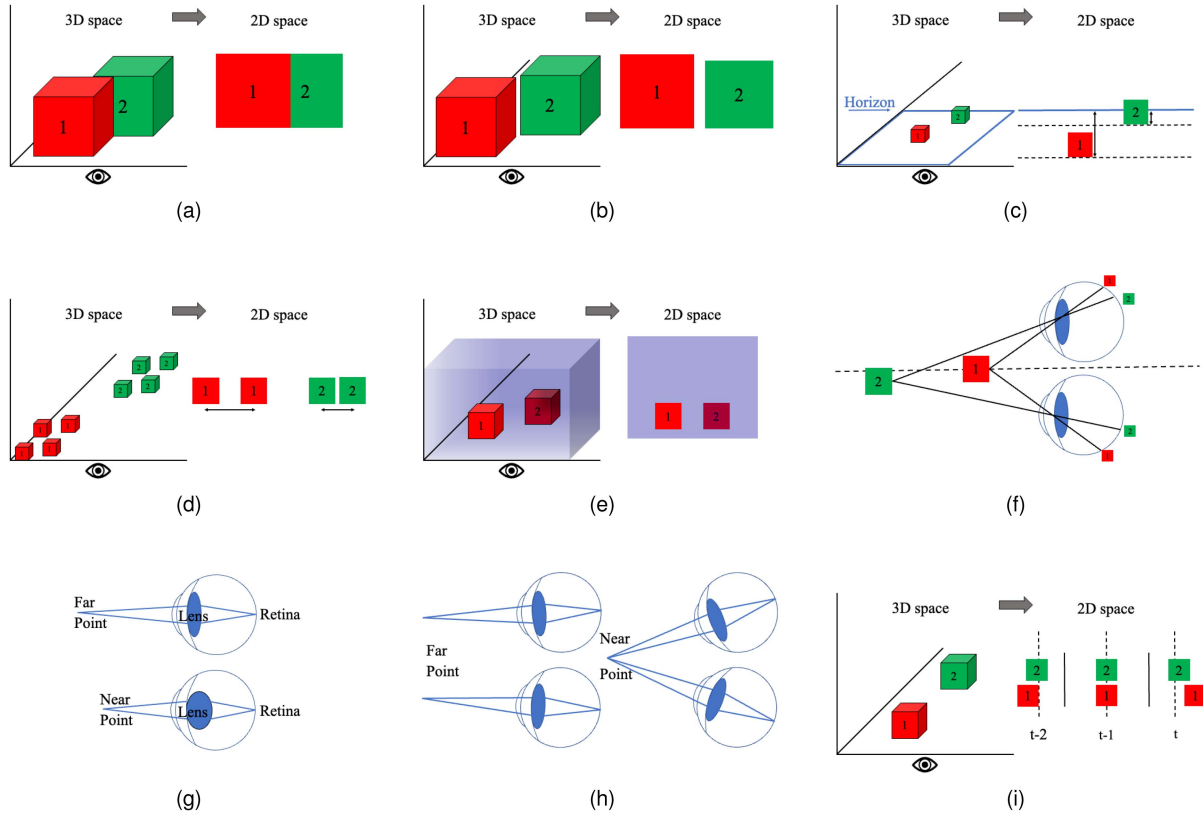


Fig. 1. Visual depth cues: (a) occlusion, (b) relative size, (c) height in the visual field, (d) relative density, (e) aerial perspective, (f) binocular disparity, (g) accommodation, (h) convergence, and (i) motion perspective. In cases (a)-(e) and (i) left side is the 3D representation of two objects and the right side is the corresponding 2D projection attained by a monocular observer.

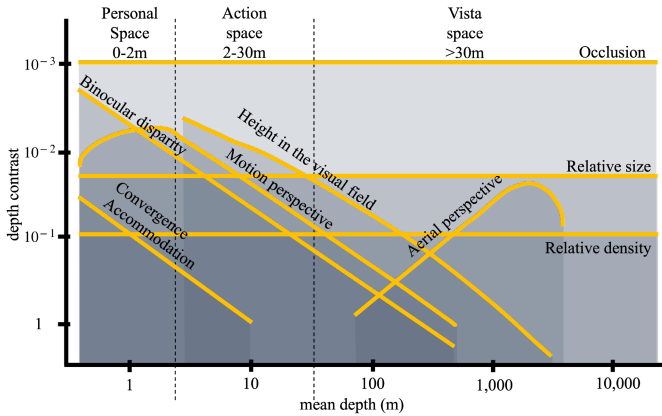


Fig. 2. Replication of Cutting and Vishton's graph on depth cues with highlighted regions of influence by each of the cues.

as texture gradients, linear perspective, brightness and shading, kinetic depth, kinetic occlusion, and gravity. According to them, these cues are “secondary” because they are either a combination of the main or they are incorrectly taken into account.

Fig. 2 is yet another reproduction of Cutting & Vishton's well-known graphical representation of the depth discrimination power versus the distance between two objects, where d_f denotes far and d_n denotes near object distance. In this graph,

the logarithmic horizontal axis is expressing the mean depth d_m defined by the two objects,

$$d_m = \frac{d_f + d_n}{2} \quad (1)$$

An additional notable peculiarity of the graph is the decreasing logarithmic vertical axis. On this axis, higher values of depth contrast are lower in the graph. Given the objects' distance, those depth cues that are below the threshold curves are effective in discriminating depth. One last peculiarity is that convergence and accommodation are considered together as suggested by Cutting & Vishton. The shading below the threshold curves in Fig. 2 emphasizes where each cue plays a role in depth discrimination. Thus, regions with shading from multiple cues are regions in which depth perception can be considered to be a multi-factor function, or depth discrimination is aided by multiple depth cues. Although, fully comprehending this graph is demanding, suffice to say that for a given mean depth (x-axis value), the region of interest for depth discrimination is upper-bounded by the cue threshold curves. Cutting & Vishton defined a practical lower limit for depth contrast at 10^{-1} . Regardless of the depth cues, estimating the depth contrast $C(d_m)$ using Cutting & Vishton's definition

$$C = 2 \frac{d_f - d_n}{d_f + d_n} \quad (2)$$

results in some interesting observations. It is clear that with increasing distances from the observer, the depth discrimination power drops significantly as the objects are close to each other. It is easy to work out the formula 2 and find out that in order for the depth contrast to be kept at a maximum, the distance of the far object should be three times that of the near object. Since this relation is constant, the further the objects are from the observer, the further they should be apart, in order for the observer to be able to discriminate their depth. As shown in Fig. 2, when the near object is at a distance of 100 m and the far object is, say, at 300 m, the mean depth is 200 m, the depth contrast is unity (1), and the depth cues that provide feedback are multiple, practically all except convergence and accommodation, which stop contributing at a mean distance of 10 m. In another case, when the near object is again at 100 m and the far object at 110.5 m, then the mean depth is 105.25 m and the contrast is 10^{-1} , which means that depth discrimination is not supported by binocular disparity, motion perspective, height in the visual field, and aerial perspective anymore.

B. Depth Estimation in 3D Digitization

Depth estimation is a basic objective of any 3D digitization method, that is, the recording of the geometric structure and, usually, also the spectral signature of 3D physical objects [33]. Initially, traditional computer vision techniques were employed, such as short-range laser beam triangulation, time-of-flight (LiDAR) scanning, phase-based scanning, shape from structured light (like the Microsoft Kinect), shape from photometry, shape from shadow, shape from silhouette, shape from shading, shape from stereo, shape from texture, depth from focus/defocus, and structure from motion (SFM). These methods utilized image correspondences, geometric relationships, and other handcrafted features to estimate depth. Some methods are monocular, others are binocular, and others rely on multiple view geometry. Detailed reviews of the various methods can be found in [33], [34]. Although 3D digitization methods reconstruct the full 3D structure of a scene, depth estimation can be limited only to the reconstruction of distances from the observer. This is the case in most recent research in monocular depth estimation. Thus, 3D digitization methods are referenced here only in an informative manner and are not discussed further.

C. Benchmark Datasets

In depth estimation research there are only a limited number of benchmark datasets freely available and widely used. These datasets have been created for various applications, mostly, though, driven by the current trend in self-driving vehicle research. What is a common requirement in depth estimation benchmark datasets is that the scene images should be accompanied by dense depth data, forming the so-called groundtruth. The depth data can be used as reference points for minimizing depth error estimates. However, most datasets are incomplete in this respect. As shown in the following paragraphs, sparse depth values are usually provided regarding their paired scenes.

In other cases, the depth maps are completely missing, as stereo images or pairs of video frames are provided for self-supervised methods. Nevertheless, researchers have been using the available benchmark datasets either to train their new methods (e.g., by using a single augmented dataset) or to estimate a new method's ability to generalize (e.g., by training on dataset A and testing on dataset B). Either way, it is not uncommon to use more than a single dataset.

Table I lists the most common datasets, reporting on the amount of data in each dataset, the data characteristics, the way the datasets were created, and their original purpose, either for indoor or outdoor applications. In addition to those listed in Table I, numerous other datasets have been developed for monocular depth estimation, including DIW [35], ETH3D [36], Sintel [37], and TUM-RGBD [38]. Researchers also combined datasets for demanding applications, like Ranftl et al. [39] who proposed a model connecting 12 datasets: ReD-Web [40], DIML [41], Movies [42], MegaDepth, WSVD [43], TartanAir [44], HR-WSI [45], ApolloSpace [46], BlendedMVS, IRS [47], KITTI, and NYU-v2.

D. Evaluation Metrics

Well-known error functions and depth accuracy metrics are used to evaluate depth map predictions. Eigen [49] was the first who recommended the most commonly used evaluation metrics for this task. Apparently, a successful method should obtain high accuracy and low error. Among a total number of pixels N , for every pixel i , the groundtruth depth is denoted as d_i and its predicted depth value as $\hat{d}_i$. The metrics used are as follows:

- *Absolute Relative Error*: $AbsRel = \frac{1}{N} \sum_{i=1}^N \frac{|d_i - \hat{d}_i|}{d_i}$
- *Squared Relative Error*: $SqRel = \frac{1}{N} \sum_{i=1}^N \frac{(d_i - \hat{d}_i)^2}{d_i}$
- *Linear Root Mean Squared Error*: $RMS E(lin) = \sqrt{\frac{1}{N} \sum_{i=1}^N (d_i - \hat{d}_i)^2}$
- *Logarithmic Root Mean Squared Error*: $RMS E(log) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\log d_i - \log \hat{d}_i)^2}$
- *Scale-invariant Mean Squared Error* [49]: $sRMS E(log) = \frac{1}{N} \sum_{i=1}^N (\log d_i - \log \hat{d}_i + a(d_i, \hat{d}_i))^2$ where $a(d_i, \hat{d}_i) = \frac{1}{N} \sum_i (\log \hat{d}_i - \log d_i)$ is the scale alignment value.
- *Accuracy with threshold* (δ_x): (%) of d_i such that $\max(\frac{d_i}{\hat{d}_i}, \frac{\hat{d}_i}{d_i}) = \delta < thr$ where $thr = 1.25, 1.25^2, 1.25^3$.

In this class of experiments, a specific accurate metric is used, the delta metric, also known as *accuracy with threshold*. This is a valuable quantitative measure that evaluates the percentage of pixels for which the ratio of correctly predicted depth values falls below a specified threshold. The calculation of δ_x enables to determine how frequently an algorithm's predictions deviate within a preselected margin from the groundtruth. By varying the threshold (typically, 1.25, 1.25², and 1.25³) we may change the strictness of the estimation, and gain insight on the tested algorithms performance. In addition, it is significant to note that, in this case, *relative metrics* are defined (AbsRel, SqRel), in that they require the normalization with the groundtruth values.

TABLE I
DATASETS MOST COMMONLY USED FOR DEPTH ESTIMATION RESEARCH

Name	Highlights	Sensor	Source
KITTI [48]	- 389 stereo and optical flow image pairs (1242×375) - stereo visual odometry sequences of 39.2 km length - more than 200k 3D object annotations - 42382 rectified stereo pairs (1242×375) - depth measurements account for 4.1% of image pixels - splits: Eigen [49], Uhrig et al. [50] - available at https://www.cvlibs.net/datasets/kitti/	LiDAR	Outdoor
NYU-v2 [51]	- 1449 RGBD monocular (640×480) - 407k monocular frames - available at https://cs.nyu.edu/~silberman/datasets/nyu_depth_v2.html	Kinect	Indoor
Make3D [52], [53], [54]	- 534 monocular (2272×1704) and their depths (55×305) - available at http://make3d.cs.cornell.edu/data.html	LiDAR	Outdoor
Cityscapes [55]	- stereo video sequences (2048×1024): 5000 with high-quality pixel-level annotations, 20000 with coarse annotations - available at https://www.cityscapes-dataset.com/	(Annotators)	Outdoor
BlendedMVS [56]	- 17,000 high-resolution images covering a wide range of scenes - available at https://github.com/YoYo000/BlendedMVS	(MVS reconstruction)	Full range
MegaDepth [57]	- 130,000 image-depth pairs - 196 locations reconstructed in 3D using COLMAP SfM/MVS - available at https://www.cs.cornell.edu/projects/megadepth/	(Internet photos)	Outdoor
DDAD [58]	- monocular videos, 150 scenes, 3080 images - 360-degree FOV, range up to 250 m, sub-1cm range precision - includes panoptic labels for 200 images - available at https://github.com/TRI-ML/DDAD	LiDAR	Outdoor
SYNS [59]	- aligned images and LiDAR panoramas from 92 scenes - dense LiDAR maps, 78.3% coverage, 365000 points per image - available at https://syms.soton.ac.uk/	LiDAR	Full range
SYNS-Patches [60]	- subset of SYNS - patches extracted at eye level in 20-degree intervals - 18 images per scene and a total dataset size of 1656 - available at https://github.com/jspenmar/monodepth_benchmark	LiDAR	Full range

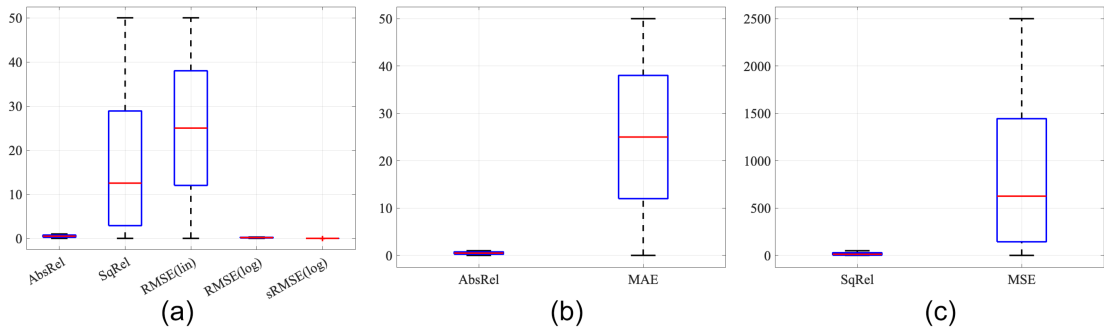


Fig. 3. Evaluation metrics: (a) AbsRel, SqRel, RMSE(lin), RMSE(log), sRMSE(log), (b) AbsRel versus MAE, (c) SqRel versus MSE.

To showcase why those metrics are preferable we use a toy experiment. Suppose we have 50 images each filled with a single gray value ranging between 50 and 100. By computing the error metrics among all possible combinations of the 50 images we get the overall statistics of the metrics, as presented in Fig. 3(a) in the form of box plots. The box plots reveal the distribution of the computed error values, and it is clear that different metrics result in totally different statistics. In addition, Fig. 3(b) and (c) show the comparison between relative and the corresponding absolute metrics, in this case, AbsRel versus its corresponding Mean

Absolute Error (MAE) and SqRel versus its corresponding Mean Squared Error (MSE). It is clear that the absolute metrics result in extremely wider error distributions, which might seem preferable. Nevertheless, the ambiguity of scale in depth estimation methods that produce predictions in an arbitrary (relative) space that is disconnected with the groundtruth space, necessitates the scaling of the error estimates to make them independent of scale, thus, usable. In a way, the usage of square roots and logarithms in RMSE(lin), RMSE(log), sRMSE(log) has a similar effect in the prediction error distributions.

It should be noted that the structural similarity index SSIM [61], the depth map edges, and the mean cosine distance between normal maps extracted from the depth images have also been used as qualitative measures of the results [62]. Furthermore, Cadena et al. [63] proposed the measure of the closest point error between the predicted depth and the groundtruth values in the 3D space, based on the Iterative Closest Point (ICP) algorithm. However, a specifically designed metric, that would be focused on the needs of the depth estimation task has not been established yet.

III. DEEP LEARNING-BASED MONOCULAR DEPTH ESTIMATION

Traditionally, depth estimation methods relied on various hypotheses, limitations, and optimizations to provide dense depth estimates. However, sensor-based approaches, like those utilizing RGB-D and LiDAR sensors, face constraints such as limited measurement range, sensitivity to outdoor lighting conditions, calibration requirements, and high energy consumption. In tasks like depth calculation with high complexity, it is nearly impossible to apply classical pattern recognition approaches, thus, the problem needs to be posed as a learning one, requiring to regress on the per-pixel depth value. In contrast, modern Deep Learning methods have achieved promising performance in estimating dense depth maps from monocular images. One way to categorize Deep Learning depth estimation methods is to consider their learning paradigms: supervised, self-supervised, and semi-supervised.

Supervised methods rely on labeled training data, where depth annotations are provided for each pixel in the training images. This allows the network to learn the mapping between the input image and its corresponding per-pixel depth values. The advantage of supervised methods is their ability to achieve high accuracy when trained on richly annotated datasets. However, acquiring large-scale labeled datasets with pixel-level depth annotations can be time-consuming and expensive.

Self-supervised methods, on the other hand, utilize the inherent structure and information present in unlabeled data to learn depth estimation. These approaches leverage image pairs or image sequences with known camera motion or scene geometry to create supervisory signals for training. By formulating the task as a reconstruction problem or a relative depth ordering problem, self-supervised methods can learn to estimate depth without explicit depth annotations. This approach alleviates the need for extensive manual labeling, but it may still require careful design choices and additional assumptions about the input data.

Semi-supervised methods aim to strike a balance between the advantages of supervised and self-supervised approaches. They leverage a combination of labeled and unlabeled data during training. By utilizing a small amount of labeled data along with a larger amount of unlabeled data, semi-supervised methods can achieve competitive performance while reducing the annotation effort compared to fully supervised methods. However, effectively leveraging both labeled and unlabeled data in a joint learning framework can be challenging and requires careful design considerations.

In recent years, various network architectures have been applied to monocular depth estimation, including Convolutional Neural Networks (CNNs), Fully Convolutional Neural Networks (FCNs), Encoders-Decoders (ED), Auto-Encoders (AEs), Generative Adversarial Networks (GANs), and more recently, Transformers. Each architecture offers unique advantages and trade-offs in terms of accuracy, efficiency, and computational requirements. In the following sections, each type of these learning modes is described in more detail.

A. Supervised Methods

The supervised methods utilize labeled images, that is, they need pairs of images and corresponding depth maps. The distance between the groundtruth and the estimated depth maps, D and $\hat{D}$ respectively, is the main characteristic of the supervised monocular depth estimation methods. Seen as a regression problem, the objective of these deep neural networks is to learn a depth map prediction rule from single images. The groundtruth depth information d_i is generally captured through sensor-based methods. Simple error metrics, such as the L_2 loss, are applied for this purpose, and should be minimized:

$$L_2(D, \hat{D}) = \frac{1}{N} \sum_{i=1}^N (d_i - \hat{d}_i)^2 \quad (3)$$

It is worth noting that while the L_2 loss is commonly used in many networks, its applicability and effectiveness may vary depending on the specific task and dataset. Therefore, its general validity as the optimal loss function for all scenarios cannot be guaranteed. However, due to its simplicity and well-established nature, many networks in the field of monocular depth estimation typically employ this loss function in their training process.

Tables II and III summarize the reviewed supervised methods in monocular depth estimation tested on the KITTI and NYU-v2 datasets respectively. The methods are sorted by the year of publication and denote with bold numbers the “winning” method in various measures and metrics¹.

Eigen et al. [49] introduced a depth prediction technique using single images. Their approach combines global and local refinements with a focus on scale-invariant error handling. Incorporating surface normals and extending depth maps to the original input resolution enhances performance. In a subsequent work, they presented a progressive refinement method across scales, preserving image details without requiring superpixels or low-level segmentation [81].

Li et al. [82] introduced a two-step approach for joint depth and surface normal estimation. The initial stage employed a multi-scale deep CNN for superpixel-level prediction. Subsequently, a pixel-level refinement was conducted using Conditional Random Fields (CRFs) [84], establishing a superpixel-to-pixel mapping via an energy function encompassing data,

¹The terminology in the Tables is as follows: *CNN*: Convolutional Neural Network; *CRF*: Conditional Random Field; *CRT*: Convolutional Regression Tree; *DP*: Depth Prediction; *ED*: Encoder-Decoder; *GL*: Global & Local; *HM*: Hilbert Maps; *SL*: Semantic Labeling; *SNE*: Surface Normal Estimation; *3DR*: 3D Reconstruction; *AbsRel*, *SqRel*, *RMSE*: Error metrics, as defined in Section II-D (lower is better); δ_1 , δ_2 , δ_3 : Accuracy metrics with thresholds, as defined in Section II-D (higher is better).

TABLE II
COMPARISON OF SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS ON KITTI DATASET

Year	Method	Approach	AbsRel ↓	SqRel ↓	RMSE ↓	RMSE log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
2022	Patil et al. [64]	Local planar information	0.071	0.270	2.842	0.103	0.953	0.993	0.998
	Yuan et al. [65]	FC-CRFs, Transformers	0.052	0.155	2.129	0.079	0.974	0.997	0.999
2021	Bhat et al. [66]	AdaBins	0.058	0.190	2.360	0.088	0.964	0.995	0.999
	Chen et al. [67]	ED, attention & ordinal loss	0.075	-	3.509	0.118	0.930	0.985	0.996
	Ranftl et al. [39]	Vision Transformers	0.096	0.739	4.479	0.062	0.915	0.977	0.993
2019	Lee et al. [68]	ED, local planar guidance	0.059	0.245	2.756	0.096	0.956	0.993	0.998
	Rosa et al. [69]	Densification, HM, Residual learning, BerHu	0.123	0.641	4.524	0.199	0.881	0.966	0.990
	Yin et al. [70]	Multi-task (DP, SNE, 3DR), ED	0.072	-	3.258	0.117	0.938	0.990	0.998
2018	Alhashim and Wonka [62]	ED	0.093	0.589	4.170	0.171	0.886	0.965	0.986
	Fu et al. [71]	Ordinal regression, CNN	0.071	0.268	2.271	0.116	0.936	0.985	0.995
	Guo et al. [72]	ED, stereo matching, occlusion mask	0.097	0.653	4.170	0.170	0.889	0.967	0.986
	Xu et al. [73]	ED & Continuous CRF, structured attention	0.122	0.897	4.677	-	0.818	0.954	0.985
2015	Liu et al. [74]	Superpixels, CNN & continuous CRF	0.217	-	7.046	0.092	0.656	0.881	0.958
2014	Eigen et al. [49]	GL prediction, scale-invariant error, CNN	0.190	1.515	7.156	0.270	0.692	0.899	0.967

TABLE III
COMPARISON OF SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS ON NYU-V2 DATASET

Year	Method	Approach	AbsRel ↓	SqRel ↓	RMSE ↓	RMSE log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
2022	Patil et al. [64]	Local planar information	0.104	-	0.356	-	0.898	0.981	0.996
	Wu et al. [75]	Transformers, stereo	0.158	-	0.548	-	0.791	0.942	0.985
	Yuan et al. [65]	FC-CRFs, Transformers	0.095	0.045	0.334	0.119	0.922	0.992	0.998
2021	Bhat et al. [66]	AdaBins	0.103	-	0.364	-	0.903	0.984	0.997
	Chen et al. [67]	ED, attention & ordinal loss	0.135	-	0.483	0.161	0.831	0.966	0.993
2020	Ranftl et al. [42]	Mixing datasets	0.100	0.061	0.407	0.057	0.905	0.984	0.997
2019	Lee et al. [68]	ED, local planar guidance	0.110	-	0.392	0.047	0.885	0.978	0.994
	Yin et al. [70]	Multi-task (DP, SNE, 3DR), ED	0.108	-	0.416	-	0.875	0.976	0.994
2018	Alhashim and Wonka [62]	ED	0.103	-	0.390	0.043	0.895	0.980	0.996
	Fu et al. [71]	Ordinal regression, CNN	0.115	0.051	0.509	-	0.828	0.965	0.992
	Xu et al. [73]	ED & Continuous CRF, structured attention	0.125	-	0.593	-	0.806	0.952	0.986
	Zhang et al. [76]	Task-attentional, BerHu loss	0.144	-	0.501	0.181	0.815	0.962	0.992
2017	Jung et al. [77]	GL prediction, GAN	0.134	0.103	0.527	-	0.822	0.971	0.993
2016	Laina et al. [78]	Residual learning, CNN, BerHu loss	0.127	-	0.573	0.195	0.811	0.953	0.988
	Mousavian et al. [79]	Multi-task (DP, SL), CNN & CRF	0.200	0.301	0.816	0.314	0.568	0.856	0.956
	Roy et al. [80]	CRT & CNN	0.187	-	-	0.744	-	-	-
2015	Eigen et al. [81]	Multi-task (DP, SNE, SL), CNN	0.158	0.121	0.641	0.214	0.769	0.950	0.988
	Li et al. [82]	Multi-task (DP, SNE), CNN & CRF	0.232	-	0.821	-	0.621	0.886	0.968
	Liu et al. [74]	Superpixels, CNN & CRF	0.213	-	0.759	-	0.650	0.906	0.976
	Wang et al. [83]	Multi-task (DP, SL), CNN & hierarchical CRF	0.220	0.210	0.745	0.262	0.605	0.890	0.970
2014	Eigen et al. [49]	GL prediction, scale-invariant error, CNN	0.215	0.212	0.907	0.285	0.611	0.887	0.971

smoothness, and auto-regression terms. Enhanced performance correlated with augmented image patches and context data. Optimal superpixel size, as determined to be 10, was emphasized. Further advancements encompassed data augmentation, deeper CNN architectures, and the potential incorporation of semantic labeling.

Liu et al. [74] employed a fusion of deep CNN and continuous CRF for monocular depth estimation. The model integrates superpixel segmentation and depth prediction through Maximum A Posteriori inference. The network comprises three components: unary and pairwise sections constituting the energy function, and a continuous CRF loss layer for negative log-likelihood minimization. An innovative addition is the superpixel pooling technique, facilitating faster training while preserving outcomes and enabling broader applicability.

Wang et al. [83] introduced a dual-layer Hierarchical Conditional Random Field (HCRF) model for indoor monocular depth and semantic prediction. The approach leverages two CNNs, one for global and another for local depth and segmentation. The global prediction CNN's loss function integrates depth and semantic label information. Conversely, the local prediction CNN, relying on over-segmentation [85], is more intricate. Training of the HCRF model minimizes the proposed energy function, yielding upper-layer image segmentation and lower-layer depth value prediction.

Laina et al. [78] introduced an efficient Fully Convolutional Residual Network (FCRN) employing modified ResNet-50 architecture with optimized up-sampling of feature maps. The method transformed RGB images (304×228 pixels) into lower-resolution depth maps (128×160 pixels). Demonstrating the

advantages of the reverse Huber (berHu) loss function [86], [87] over L_2 , they also assessed the technique's accuracy in a Simultaneous Localization and Mapping context.

Mousavian et al. [79] addressed concurrent semantic segmentation and depth estimation using a multi-scale deep CNN. A single loss function, encompassing Eigen's scale-invariant loss [49] and semantic segmentation loss, facilitates simultaneous learning. Performance enhancement is achieved through the incorporation of a fully connected CRF module.

Roy et al. [80] proposed Neural Regression Forests (NRF) merging binary Convolutional Regression Tree (CRT) nodes with a shallow Convolutional Neural Network. Images are analyzed in parallel with patches forwarded to CRT child nodes based on Bernoulli probability. CRT leaves serve as weak depth estimators, complemented by standard bilateral filtering [88] to ensure depth prediction smoothness.

Jung et al. [77] introduced a Generative Adversarial Network (GAN) approach to monocular depth estimation, comprising a generator and discriminator network. The generator comprises global and refinement segments: a fully convolutional encoder-decoder predicts global depth, while local depths are estimated via image patches. The discriminator is active during training, assessing prediction authenticity. After min-max training, the generator produces faithful depth images, marked by their likeness to ground truth.

Alhashim and Wonka [62] proposed an encoder-decoder network with skip connections for high-resolution (320×240) depth estimation from RGB images (640×480). The encoder, pretrained on DenseNet-169 [89], and decoder with convolutional layers and bilinear upsampling blocks form the architecture. The approach emphasizes controlled downsampling during encoder pre-training to preserve spatial information. They introduced a novel loss function incorporating point-wise L1 loss, image gradient, and SSIM [61]. Method robustness extended to Unreal-1 k dataset, with data augmentation involving horizontal flipping and color channel swapping.

Fu et al. [71] introduced a Deep Ordinal Regression Network (DORN) for high-resolution depth maps, incorporating a Spacing-Increasing Discretization (SID) strategy and ordinal regression training loss. The network architecture integrates dense feature extraction, a multi-scale feature learner (ASPP), cross-channel information learning, a full-image encoder, and ordinal regression optimization. DORN's versatility is demonstrated through successful application to indoor and outdoor images.

Guo et al. [72] introduced a unique pipeline for monocular depth estimation, overcoming challenges of supervised and unsupervised methods. They employ a deep stereo-matching network to learn depth from synthetic data, offering dense supervision for training a monocular depth estimator. The pipeline features strategies enhancing proxy stereo network fine-tuning, resulting in sharper boundaries, enhanced occlusion prediction, and improved overall performance.

Xu et al. [73] incorporated a Structured Attention guided CRF model between their encoder-decoder setup. The attention module governs features across encoder scales, employing

ResNet-50 [90] for CNN components. The CRF model employs a three-term energy function amalgamating multi-scale feature maps, estimated representations, and attention variables.

Zhang et al. [76] presented Task-Recursive Learning, a model predicting depth and semantic labels. ResNet-50 [90] constitutes the initial part, followed by concatenated Task-Attentional Modules and convolutional layers, enabling depth and semantic predictions at varying scales.

Lee et al. [68] introduced an enhancement to the decoder stage in an encoder-decoder setup, employing local planar guidance layers for sharper depth boundaries. The approach incorporates dense feature extraction, contextual information extraction (ASPP), and densely connected local planar guidance layers. While boundary sharpness improved, issues in upper scene areas and the need for added photometric reconstruction were detected.

Rosa et al. [69] addressed depth map sparsity by proposing a supervised framework that utilizes the Hilbert Maps approach [91] for densification in monocular depth estimation. The FCRN model, inspired by Laina et al. [78], employs residual upsampling blocks in lieu of fully-connected layers. While not necessitating additional data, the method incurs extra computational costs per continuous depth map during training, with results influenced by Hilbert Maps' reconstruction capabilities. Noteworthy enhancements encompass architecture optimization, supplementary techniques, and color distortion utilization.

Yin et al. [70] presented a supervised approach calculating high-order 3D geometric features post-depth prediction and 3D point cloud reconstruction. Employing an encoder-decoder network for depth prediction, they introduced a loss term named Virtual Norm (VN) resilient to depth noise. The VN term highlighted the value of incorporating supplementary 3D geometry information in depth estimation.

Ranftl et al. [42] introduced methods enabling diverse dataset fusion for comprehensive monocular depth estimation. They advocate principled multi-objective learning, promoting robustness across depth range and scale variations. Emphasis lies on encoder pre-training for additional tasks. Utilizing five training datasets and 3D film data, they extended their approach [39] by incorporating five more datasets. The architecture, named "dense prediction transformer," was developed and contrasted against fully-convolutional models.

Aich et al. [92] proposed a Bidirectional Attention Network (BANet) as an end-to-end framework for monocular depth estimation. Their approach tackled the challenge of integrating local and global information in convolutional neural networks, by leveraging insights from neural machine translation. As a result, they developed a lightweight mechanism for adaptive computation control, similar to the dynamic nature of recurrent neural networks. Bidirectional attention modules were introduced, leveraging feed-forward feature maps to incorporate global context and mitigate ambiguity.

Bhat et al. [66] introduced AdaBins, a building block based on transformers. AdaBins partitions the predicted depth range into bins determined adaptively and computes the ultimate depth values by linearly combining the bin centers. By employing

a Transformer encoder as a building block after a CNN, the approach aims to incorporate non-local processing and achieve a comprehensive understanding of the scene.

Chen et al. [67] introduced ACAN, an Attention-based Context Aggregation Network, addressing dense labeling in monocular depth estimation. Comprising an encoder, decoder (Context Aggregation Module - CAM), and soft ordinal inference module, the network leverages ResNet architecture [90] for dense feature extraction. By collecting discriminative image and pixel-level context information, CAM refines predictions. Attention and ordinal losses drive model training, while soft ordinal inference converts probabilities to continuous depth values. The approach mitigates grid artifacts observed in prior methods like ASPP-based approaches [93], [94], as shown in results on cropped and resized images.

Miangoleh et al. [95] focused on the impact of input resolution and scene structure on depth estimation performance, identifying a trade-off between consistent scene structure and high-frequency details. To address this, the authors proposed a method that combines low- and high-resolution estimations, generated by a pre-trained model [42]. The low-resolution estimations obtained a base estimate with a consistent structure, while the incorporated patch-level (high-frequency) estimates enhanced local details. The authors also demonstrated the effectiveness of their approach in boundary localization through simple segmentation using depth thresholding.

Qiao et al. [96] presented ViP-DeepLab, a unified model designed to address the challenging inverse projection problem in vision, which involved restoring point clouds from perspective image sequences while providing instance-level semantic interpretations for each point. Referred to as Depth-aware Video Panoptic Segmentation, this joint task combined monocular depth estimation and video panoptic segmentation. The authors introduced a novel evaluation metric, Depth-aware Video Panoptic Quality (DVPQ), and provide two derived datasets to support future research in this area.

Yin et al. [97] tackled single-image monocular 3D scene shape recovery, addressing depth shift and camera focal length uncertainties. Their two-stage framework initially predicts depth up to an unknown scale and shift, followed by 3D point cloud encoders estimating the missing depth shift and focal length. They introduce image-level normalized regression and normal-based geometry losses to enhance depth prediction models trained on mixed datasets.

Patil et al. [64] introduced P3Depth, a supervised method that incorporates a piecewise planarity prior, emphasizing the effectiveness of integrating geometric priors for accurate depth estimation. The authors leverage the high regularity of real 3D scenes to selectively utilize information from coplanar pixels, aiming to improve the accuracy of depth predictions. Their proposed network architecture consists of two heads: one outputting pixel-level plane coefficients and the other generating a dense offset vector field to identify seed pixels. By utilizing the plane coefficients of seed pixels, depth predictions are made for each position. To account for potential deviations from precise local planarity, the initial predictions are adaptively fused with the

plane coefficient-based predictions using a learned confidence measure.

Wu et al. [75] proposed DistDepth, a structure distillation approach for enhancing monocular depth estimation accuracy through self-supervised learning. By leveraging an off-the-shelf relative depth estimator [39] and left-right consistency, DistDepth distills depth-domain structure knowledge to obtain depth maps without curated depth annotations. The framework focused on complex indoor scenes by combining structure distillation with a branch that learns metrics. The authors collected SimSIN, a large-scale indoor simulation dataset, and UniSIN, a real dataset, to facilitate learning and evaluation.

Yuan et al. [65] explored the Conditional Random Fields (CRFs) optimization, introduced a window-based approach by splitting the input image into sub-windows and performing optimization within each window. This window-based strategy reduces the computation complexity while making fully-connected CRFs feasible. To capture the relationships between graph nodes effectively, the authors employed a multi-head attention mechanism to compute a multi-head potential function. This neural CRFs module serves as the decoder in their proposed bottom-up-top-down network architecture, while a vision transformer was employed as the encoder.

The main advantage of supervised depth estimation is its performance compared to the other learning type methods. However, the groundtruth depth information is hard to obtain. As a result, the scope of the supervised depth estimation methods is limited by the availability of groundtruth data.

B. Self-Supervised Methods

Since depth information is not always available in designing learning methods, in many cases the unsupervised/self-supervised methods are being employed. Such methods produce the supervisory signal by themselves. Usually, they use data from stereo images or video frames. As a result, the term “unsupervised” is not clear in the related bibliography and the term “self-supervised” is more accurate. The training process scope is the learning of the image reconstruction rule among the images, guided by a loss function. Monocular image sequences can be exploited to extract the geometric constraints from frames. Let p_t denote the homogeneous coordinates of a pixel in the target view I_t . Then, p_t is projected onto the source view I_s as p_s :

$$p_s \sim \mathbf{K} \mathbf{T}_{t \rightarrow s} D_t(p_t) \mathbf{K}^{-1} p_t \quad (4)$$

where $\mathbf{K}$ is the known camera intrinsics matrix, $D_t(p_t)$ is the depth value at p_t , and $\mathbf{T}_{t \rightarrow s}$ is the spatial transformation matrix from I_t to I_s . Note that $D_t(p_t)$ and $\mathbf{T}_{t \rightarrow s}$ can be also estimated when they are not given, meaning that

$$p_s \sim \mathbf{K} \hat{\mathbf{T}}_{t \rightarrow s} \hat{D}_t(p_t) \mathbf{K}^{-1} p_t \quad (5)$$

For example, Zhou et al. [98] used a single-view CNN to estimate $\hat{D}_t(p_t)$ and a pose network to estimate $\hat{\mathbf{T}}_{t \rightarrow s}$. The training of such neural networks is based on photometric error. In other words, given a targeted view I_t , captured at time t , and its N nearby source views I_s ($1 \leq s \leq N$, $s \neq t$), the objective

is the minimization of the photometric loss, also known as the view synthesis objective:

$$L_{vs} = \sum_s \sum_p |I_t(p_t) - \hat{I}_s(p_s)| \quad (6)$$

Zhou et al. [98] proposed that the synthesized view, $\hat{I}_s(p_s)$, can be estimated in a differentiable manner, through bilinear interpolation of 4 neighbors of the projected coordinate p_s . As the gradient signals should be very local, they proposed a multi-scale prediction mechanism. In that way, the gradients are derived from a larger region in higher scales and gradually propagated to lower scales.

The loss function can also be enhanced with the Structural Similarity Index (SSIM) [61], known as the rigid warping loss [99]:

$$L_{rw} = \sum_s \alpha \frac{1 - SSIM(I_t, \hat{I}_s)}{2} + (1 - \alpha) |I_t - \hat{I}_s| \quad (7)$$

where α is a balance factor.

Furthermore, the erroneous predictions can be filtered out and the sharp details are preserved by the edge-aware depth smoothness loss [99], [100], [101]:

$$L_{ds} = \sum_p |\nabla D(p)| \cdot (e^{-|\nabla I(p)|})^\tau \quad (8)$$

where p denotes homogeneous coordinates of pixels in frame I and D_p corresponding depth value, ∇ is the vector differential operator, $|\cdot|$ denotes the element-wise absolute value.

Due to scale ambiguity, scale inconsistency, and occlusion problems, methods based on synchronized image pairs have been proposed. Without groundtruth depth data, the learning procedure is based on left-right image pairs. The goal of this type of neural network is to learn the disparity maps, which represent the horizontal shift between corresponding pixels in the stereo pair, providing information about the depth or 3D structure of the scene. The training loss is based on the comparison between the original I and the synthesized images $\hat{I}$, leveraging the learned disparity information:

$$L_{recon} = \sum_p \left\| I_l(p) - \hat{I}_l(p) \right\|^2 = \sum_p \left\| I_l(p) - I_r(p + d(p)) \right\|^2 \quad (9)$$

where I_l , I_r are the original left-right images respectively and $\hat{I}_l$ is the reconstructed left image based on the original right image.

Table IV summarizes the reviewed methods for unsupervised/self-supervised methods in monocular depth estimation. The methods are sorted by the year of publication and denote with bold numbers the “winning” method in various measures and metrics².

²In this Table, the terminology is as follows: *CNN*: Convolutional Neural Network; *CS*: Cityscapes dataset; *DVO*: Direct Visual Odometry; *GAN*: Generative Adversarial Network; *ICP*: Iterative Closest Point; *LSTM*: Long Short-Term Memory; *AbsRel*, *SqRel*, *RMSE*: Error metrics, as defined in Section II-D (lower is better); δ_1 , δ_2 , δ_3 : Accuracy metrics with thresholds, as defined in Section II-D (higher is better).

1) *Image Sequences*: Zhou et al. [98] employed a CNN-based approach integrating a monocular depth network and multi-view pose network. The method features warping-based view synthesis and permits independent network operation during testing. However, the approach overlooks scene dynamics, occlusions, and treats camera intrinsics as fixed inputs.

Kumar et al. [124] introduced a GAN-based approach for depth and pose estimation using video sequences. The generator comprises depth and pose networks, predicting depth maps and rotation/translation parameters for consecutive frames. Employing [98] inspiration, depth is predicted from the current image, and pose is estimated from a triplet of previous, current, and next frames. Depth maps and pose parameters form image pairs following [143]. The discriminator distinguishes real from generated images, leading to a three-term loss function involving photo-consistency, adversarial, and depth terms for network training.

Mahjourian et al. [125] introduced an unsupervised approach for depth and ego-motion estimation. Their method incorporates a 3D loss term based on the Iterative Closest Point (ICP) method [144], [145], [146], encompassing the 3D scene geometry. The loss function additionally integrates photometric consistency, structured similarity, and smoothness terms. Principled occlusion and disocclusion masks handle adjacent frame inconsistencies. Generalization to uncalibrated mobile phone camera data was also evaluated.

Wang et al. [122] introduced a single-image depth network and a video frame ego-motion estimation pose network. The approach incorporates masks to account for occlusion pixels between frames, eliminating erroneous matching pixels. This information contributes to the proposed loss function terms.

Yin and Sin [99] unified depth, optical flow, and ego-motion in one system. They divided static and dynamic objects into two sub-networks: the first, named rigid structure reconstructor, comprises depth and camera motion modules; the second, named non-rigid motion localizer, generates the final flow map. Geometric consistency enforcement addresses occlusions and texture ambiguities in the last step.

Zou et al. [127] co-trained depth, pose, and optical flow estimation networks capable of independent operation during testing. While inspired by Zhou et al. [98] architecture, the introduction of cross-task consistency loss demonstrated performance enhancement. Notably, optical flow predictions between flow estimation, depth, and pose networks achieved consistency. The total loss function integrates photometric, smoothness, forward-backward, and cross-consistency terms.

Almalioglu et al. [115] introduced an adversarial learning approach for simultaneous pose and depth estimation using video sequences. The method employs a multi-view pose regressor and a single-view depth generator feeding into a view reconstruction module, generating synthetic images. The discriminator distinguishes between the reconstructed and original images, training halting when close similarity is achieved. The pose estimation module combines a CNN and Recurrent network, while depth estimation necessitates an encoder before the generator network.

TABLE IV
COMPARISON OF UNSUPERVISED/SELF-SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS ON THE KITTI DATASET

Year	Method	Approach	AbsRel ↓	SqRel ↓	RMSE ↓	RMSE _{log} ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
image sequences	Guizilini et al. [102]	Multi-frame, transformers, cost volume.	0.090	0.661	4.149	0.175	0.905	0.967	0.984
	2022 Han et al. [103]	Transformers, PWSA module, self-distill loss	0.095	0.711	4.321	0.172	0.906	0.967	0.984
	Ma et al. [104]	Attention module, non-boundary triplet loss	0.102	0.656	4.339	0.175	0.892	0.967	0.985
	Zhao et al. [105]	Transformers & Convolutional layers	0.099	0.708	4.372	0.175	0.900	0.967	0.984
	2021	Lyu et al. [106]	0.109	0.792	4.632	0.185	0.884	0.962	0.983
		Ruhkamp et al. [107]	0.103	0.746	4.483	0.180	0.894	0.965	0.983
		Sun et al. [108]	0.117	0.863	4.813	0.192	0.871	0.959	0.982
		Yan et al. [109]	0.105	0.769	4.535	0.181	0.892	0.964	0.983
		Watson et al. [110]	0.090	0.713	4.261	0.170	0.914	0.966	0.983
		Zhou et al. [111]	0.102	0.764	4.483	0.180	0.896	0.965	0.983
	2020	Johnston et al. [112]	0.106	0.861	4.699	0.185	0.889	0.962	0.982
		Klingner et al. [113]	0.113	0.835	4.693	0.191	0.879	0.961	0.981
		Tosi et al. [114]	0.120	0.792	4.750	0.191	0.856	0.958	0.984
	2019	Almalioglu et al. [115]	0.150	1.141	5.448	0.216	0.808	0.939	0.975
		Bian et al. [116]	0.137	1.089	5.439	0.217	0.830	0.942	0.975
		Casser et al. [101]	0.109	0.825	4.750	0.187	0.874	0.958	0.983
		Chen et al. [117]	0.100	0.811	4.806	0.189	0.875	0.958	0.982
		Gordon et al. [118]	0.128	0.959	5.230	-	-	-	-
		Guizilini et al. [58]	0.078	0.420	3.485	0.121	0.931	0.986	0.996
		Li et al. [119]	0.150	1.127	5.564	0.229	0.823	0.936	0.974
		Ranjan et al. [120]	0.140	1.070	5.326	0.217	0.826	0.941	0.975
		Sun et al. [121]	0.156	0.948	4.462	0.227	0.789	0.934	0.976
		Wang et al. [122]	0.158	1.277	5.858	0.233	0.785	0.929	0.973
		Wu et al. [123]	0.063	0.178	2.129	0.097	0.961	0.993	0.998
	2018	Kumar et al. [124]	0.120	0.747	4.756	0.187	0.849	0.955	0.985
		Mahjourian et al. [125]	0.163	1.240	5.912	0.243	0.784	0.923	0.970
		Wang et al. [126]	0.151	1.257	5.583	0.228	0.810	0.936	0.974
		Yin and Sin [99]	0.155	1.296	5.857	0.233	0.793	0.931	0.973
		Zou et al. [127]	0.150	1.124	5.507	0.223	0.806	0.933	0.973
2017	Zhou et al. [98]	Monocular framework, mask network	0.208	1.768	6.856	0.283	0.678	0.885	0.957
stereo images	2021	Choi et al. [128]	0.098	0.621	4.215	0.185	0.885	0.961	0.982
		Peng et al. [129]	0.091	0.646	4.207	0.176	0.901	0.966	0.983
	2019	Andraghetti et al. [130]	0.091	0.548	3.690	0.181	0.892	0.956	0.979
		Feng et al. [131]	0.065	0.673	4.003	0.136	0.944	0.979	0.991
		Godard et al. [132]	0.115	0.903	4.863	0.193	0.877	0.959	0.981
		Goldman et al. [133]	0.169	0.653	3.790	0.195	0.867	0.954	0.979
		Pilzer et al. [134]	0.098	0.831	4.656	0.202	0.882	0.948	0.973
		Tosi et al. [135]	0.111	0.867	4.714	0.199	0.864	0.954	0.979
		Wang et al. [136]	0.049	0.515	3.404	0.121	0.965	0.984	0.992
		Watson et al. [137]	0.106	0.780	4.695	0.193	0.875	0.958	0.980
	2018	Aleotti et al. [138]	0.119	1.239	5.998	0.212	0.846	0.940	0.976
		Luo et al. [139]	0.094	0.626	4.252	0.177	0.891	0.965	0.984
		Luo et al. [139]	0.094	0.626	4.252	0.177	0.891	0.965	0.984
		Pilzer et al. [140]	0.152	1.388	6.016	0.247	0.789	0.918	0.965
		Poggi et al. [141]	0.126	0.961	5.205	0.220	0.835	0.941	0.974
2017	Godard et al. [142]	Left-right consistency loss	0.148	1.344	5.927	0.247	0.803	0.922	0.964
2016	Garg et al. [14]	Stereo framework	0.169	1.080	5.104	0.273	0.740	0.904	0.962

Bian et al. [116] achieved globally scale-consistent depth prediction in long video sequences by utilizing geometry consistency loss. Their method handles moving objects and occlusions through self-discovered masks. Depth and pose networks play pivotal roles, with DispResNet [120] exhibiting superior performance in depth estimation over DispNet [98]. PoseNet [98] was used for the pose network.

Casser et al. [101] addressed monocular depth and ego-motion via video sequences. Building on Zhou et al. [98] and Godard et al. [132], their algorithm enhances motion estimation, ego-motion, and introduces online refinement for unknown data performance. Training mandates a minimum of three images for input into the encoder-decoder depth network.

Chen et al. [117] used depth estimation, optical flow, and camera pose neural networks for multitasking in monocular

videos. The contribution lies in the adaptive photometric consistency and geometric constraints encompassed in the loss function. The method additionally predicts camera intrinsics in the absence of explicit information. Improved generalization to new environments was achieved through the combination of two online refinement processes.

Gordon et al. [118] introduced two Unet-based networks. The first estimates single-image depth, while the second predicts ego-motion, object motion, and camera intrinsic parameters using two image frames. Both networks train jointly, and the method includes an occlusion-aware consistency loss as a key contribution.

Guizilini et al. [58] trained their PackNet deep network using unlabeled monocular videos and introduced the Dense Depth for Automated Driving (DDAD) dataset with extensive 200 m

range depth maps. The encoder-decoder architecture features packing-unpacking blocks, skip connections [147], and a scale-aware Structure from Motion (SfM) training approach incorporating appearance matching, depth smoothness, and velocity supervision losses.

Li et al. [119] introduced a sequential approach for depth estimation and visual odometry. Optical flow [148] encodes past frame information compactly, managed by an LSTM [149] in the generator module for depth and pose estimations. A discriminator distinguishes fake from real images, and training loss combines four components: appearance, depth regularization, trajectory consistency, and GAN loss.

Ranjan et al. [120] addressed depth estimation, camera motion, optical flow, and motion segmentation challenges via Competitive Collaboration. Sub-networks, including competitors for depth and camera motion and a moderator for motion segmentation, mutually reinforce through competition and cooperation. In the first phase, the moderator guides competitors; the second phase involves joint training. The proposed loss function combines individual losses for coherent sub-network training.

Sun et al. [121] incorporated depth and camera motion estimation inspired by Zhou et al. [98], Godard et al. [142], and introduced novel terms to the cost function. Their function emphasizes bidirectional view reconstruction consistency by minimizing projection photometric differences.

Wang et al. [126] integrated Direct Visual Odometry (DVO) [150] into their approach for monocular depth estimation from video sequences, addressing the scale ambiguity inherent in such methods through a proposed normalization strategy. They replaced a traditional pose prediction network with a differentiable implementation of DVO, resulting in enhanced depth estimation performance.

Wu et al. [123] incorporated a spatial correspondence module into a generative adversarial network designed for video-based depth estimation, inspired by the Flownet framework [151]. They utilized Smolyak's sparse grids [152] to efficiently down-sample and compare feature maps, achieving effective spatial correspondence. To account for various directions, they introduced a direction-based attention mechanism, similar to the approaches presented in DSC (Directional Spatial Context) [153] and Squeeze-and-Excitation Blocks [154]. Furthermore, the depth map resolution was enhanced through a dedicated refinement sub-network.

Guizilini et al. [155] proposed an approach that leverages semantic information to guide the generation of depth features. The architecture incorporates a fixed pretrained semantic segmentation network to facilitate semantic-dependent geometric representation learning through pixel-adaptive convolutions. The authors also introduced a two-stage training process to address biases on dynamic objects and achieve more accurate predictions.

Johnston et al. [112] proposed an approach that combines self-attention and discrete disparity prediction. The self-attention mechanism allows the model to capture contextual information from non-contiguous image regions, while the discrete disparity prediction enhances depth estimation sharpness, the model robustness, and enables depth uncertainty estimation.

The architecture includes a self-attention context module, pose estimator, and multi-scale decoder.

Klingner et al. [113] proposed SGDepth, a method that addresses the challenge of moving dynamic-class (DC) objects in the scene. To overcome the limitations of static-world assumptions during training, the authors introduced a cross-domain training approach that combines supervised semantic segmentation and self-supervised depth estimation. A semantic masking scheme was implemented to prevent moving DC objects from affecting the photometric loss. Additionally, a detection method was introduced to identify frames with non-moving DC objects, enabling the learning of depth for such objects.

Tosi et al. [114] presented a comprehensive framework for scene understanding from monocular videos, encompassing depth estimation, optical flow, semantics, and motion segmentation. Their novel network architecture, Ω Net, enables real-time prediction on both high-end GPUs and low-power embedded platforms. The authors introduced a training protocol that leverages self-supervision, knowledge distillation, and multi-task learning, with proxy semantic labels and self-distillation playing key roles.

Lyu et al. [106] addressed the limitations of high-resolution monocular depth estimation and proposed HR-Depth, an improved U-net architecture. The authors identified inaccurate depth estimation in large gradient regions as a core issue and introduced two effective strategies to enhance performance. First, they re-designed the skip-connections to obtain high-resolution semantic feature maps. Second, a feature fusion Squeeze-and-Excitation (fSE) module was proposed to improve the efficiency and effectiveness of feature fusion. HR-Depth is available in two versions: one utilizing ResNet-18 [90] and another lightweight version with MobileNetV3 [156] as the encoder.

Ruhkamp et al. [107] introduced the TC-Depth network, which utilizes a transformer architecture for depth estimation. The network includes a spatial attention module that correlates coarse depth predictions to capture local geometric information and a temporal attention mechanism that processes this information across consecutive frames. They also proposed cycle consistency regularization to enhance depth consistency. Furthermore, the authors introduced a temporal consistency metric (TCM) to objectively evaluate depth consistency across frames, offering a valuable criterion for future research in this field.

Sun et al. [108] proposed an approach using hybrid masks for monocular depth and visual odometry (VO) estimation in dynamic environments. The authors introduced two types of masks: the cover mask and the filter mask. The cover mask focuses on the accuracy of the VO estimation by masking potential outliers, while the filter mask filters out outliers during view reconstruction. Additionally, a depth-pose consistency loss was introduced to address scale inconsistency between different training samples.

Yan et al. [109] introduced CADepth-Net, a network incorporating two innovative modules: the Structure Perception Module, which employs self-attention to capture long-range dependencies and enhances scene understanding, and the Detail Emphasis Module, which recalibrates feature maps to selectively emphasize informative features for improved depth prediction.

precision. The framework combines a fully convolutional U-Net architecture with ResNet encoders, structure perception modules, and detail emphasis modules to achieve sharper depth estimation results.

Watson et al. [110] proposed ManyDepth, an adaptive approach to self-supervised monocular depth estimation that leverages sequence information at test time. Unlike existing methods that either ignore the sequence information or rely on computationally expensive test-time refinement or recurrent networks, ManyDepth utilizes a deep end-to-end cost volume-based approach. The network is trained using self-supervision and incorporates a novel consistency loss to disregard unreliable depth estimations, such as those affected by moving objects. Additionally, an augmentation scheme was introduced to handle static cameras.

Zhou et al. [111] introduced DIFFNet, an encoder-decoder network derived from the HRNet semantic segmentation model [157]. DIFFNet incorporates feature fusion and an attention mechanism within the decoder to leverage the semantic information effectively during the downsampling and upsampling stages.

Guizilini et al. [102] proposed DepthFormer, a multi-frame method for self-supervised monocular depth estimation. The key contribution is a cross-attention module that improves feature matching between images by addressing the limitations of traditional appearance-based similarity metrics. Depth-discretized epipolar sampling is used to select matching candidates, and a series of self- and cross-attention layers sharpen the matching probability, overcoming ambiguities and local minima. The refined cost volume is then decoded into depth estimates, and the entire pipeline is trained end-to-end using only a photometric objective.

Han et al. [103] introduced TransDSSL, an approach that combines transformers and self-supervised learning for monocular depth estimation. By leveraging the capability of transformers to encode long-range spatial dependencies, the proposed model addresses the challenges associated with adopting transformers in depth estimation tasks. The authors propose the Pixel-Wise Skip Attention (PWSA) module, which enhances fine details in feature maps while preserving global context from the transformer encoder. Additionally, a self-distillation loss is introduced to improve the training stability and enhance intermediate representations.

Ma et al. [104] addressed the underappreciated distinctions between depth estimation and semantic segmentation tasks. Their solution involves an attention-based module to enhance task-specific features while preserving cross-task spatial coherence. Additionally, they introduce a metric learning-based approach to separate instances in the feature space. The pipeline features DepthSegNet and PoseNet components, with the novel intra/cross-task multi-head attention module placed between task-specific decoders. Training loss includes photometric, smoothness, boundary triplet, semantic cross-entropy, and the proposed “non-boundary triplet” losses.

Zhao et al. [105] proposed MonoViT, a framework for self-supervised monocular depth estimation that combines CNNs and Vision Transformers (ViTs). By integrating Transformer

blocks into the backbone along with convolutional layers, MonoViT achieves both local and global reasoning, surpassing the limitations of CNNs with restricted receptive fields. Simultaneously, a lightweight PoseNet is utilized for pose estimation between temporally adjacent images.

Poggi et al. [158] presented a comprehensive study on uncertainty estimation in self-supervised monocular depth estimation, addressing an important yet unexplored aspect of this field. By evaluating various uncertainty estimation approaches and investigating the impact of different self-supervised training paradigms, the authors demonstrated the significance of modeling uncertainty for practical applications. They introduced a novel Self-Teaching paradigm that improves depth accuracy, particularly when the pose is unknown during training.

2) *Stereo Images*: Garg et al. [14] introduced an early unsupervised depth estimation method for stereo images. The approach also applies to image pairs with minor known camera motion. The architecture consists of an auto-encoder. The encoder, a CNN with stacked convolutions and pooling, predicts depth from the left image. The decoder employs inverse warping, where the right image and depth map produce a warped image, minimizing the reconstruction error with the left image. Experimental results demonstrated enhanced depth predictions through the proposed upsampling.

Godard et al. [142] presented an encoder-decoder network deriving disparities from monocular images through left-to-right and right-to-left calculations. The approach introduces a left-right consistency term alongside smoothness and reconstruction terms in the loss function. Later, Godard et al. [132] proposed enhancements for monocular depth estimation from video or stereo data. These include per-pixel minimum photometric error for occlusion handling, auto-masking for detecting stationary pixels, and a full-resolution multi-scale upsampling technique to reduce artifacts.

Alleoti et al. [138] introduced an early instance of Generative Adversarial Networks (GANs) for depth estimation using stereo image pairs. Their GAN comprises a generator producing new images and a discriminator trained to differentiate real from generated (warped) images. These networks engage in a min-max game controlled by an objective function.

Luo et al. [139] decomposed monocular depth estimation into view synthesis and stereo matching sub-problems. Their approach involves a view synthesis network generating a synthesized right image from the original left image, followed by correlations computation and depth map derivation using a stereo-matching network. DispNet [159] handles correlations, while DispFullNet [160] generates full-resolution depth maps.

Pilzer et al. [140] introduced cyclic generative networks for monocular depth estimation. Two symmetric sub-networks synthesize images from multiple views, combining information for depth prediction. The first generator uses left and right images to predict disparities, enhancing them to compute disparity maps and synthesize right-view images. The second generator follows the same process using synthesized images and the original left image. Discriminators distinguish original from synthesized inputs. Training loss includes reconstruction, adversarial, and consistency (disparity map similarity) terms.

Poggi et al. [141] leveraged trinocular assumptions in binocular images by estimating the reference central image. Their method optimizes disparities between central-left, left-central, central-right, and right-central images, requiring collaboration between two encoder-decoder sub-networks.

Andraghetti et al. [130] enhanced monocular depth estimation using sparse depth points initially obtained via stereo visual odometry. These points were densified using an auto-encoder and incorporated into existing depth estimation methods by Godard et al. [142] and Poggi et al. [161].

Feng et al. [131] introduced an unsupervised neural network for depth and ego-motion estimation using stacked GAN layers and video sequences. The bottom layer handles depth and ego-motion estimation at each time step, while upper layers capture spatial and temporal dynamic features. Each layer includes generator-discriminator units contributing to the corresponding loss. The method requires both left and right images of the current frame for training, with the final loss being a weighted sum across layers.

Goldman et al. [133] introduced a siamese network employing shared weights for self-supervised monocular depth estimation, similar to Godard et al. [142]. Each subnetwork predicts disparity maps from single left or right images using a defined loss function. Built upon DispNet architecture [159], ResNet [90], and VGG variants [166], their encoder-decoder network with skip connections demands time and memory during training due to symmetric data use. Nonetheless, only one subnetwork is utilized in the testing phase.

Pilzer et al. [134] introduced a three-sub-network monocular depth estimation approach. The first sub-network generates right-to-left disparity and synthesizes the left image, while the second predicts left-to-right disparity for right image re-synthesis. Leveraging inconsistencies between original and reconstructed right images, the third sub-network refines the depth map. Knowledge distillation transfers learning from the third sub-network to the first, enhancing depth prediction performance.

Tosi et al. [135] introduced monoResMatch, a self-supervised end-to-end network. Comprising a multi-scale feature extractor, initial disparity estimation, and refinement, it employs image and synthesized features aligned with a virtual right image for depth estimation. Stereo matching, using Semi-Global Matching (SGM) [167], generates accurate proxy labels for the final left-view aligned disparity map.

Wang et al. [136] unified optical flow and stereo depth estimation using three lightweight CNNs concurrently. Stereo images feed the depth network, while video frames drive the camera ego-motion and optical flow networks. Camera motion refinement employs the rigid-aware direct visual odometry module, which computes a rigid mask, affecting the loss function terms.

Watson et al. [137] introduced Depth Hints to overcome local minima in self-supervised monocular depth estimation. These hints enhance the photometric loss function and improve the performance of training algorithms, particularly for capturing thin structures and sharp boundaries.

Choi et al. [128] proposed a novel framework for monocular depth estimation that leverages pseudo-depth labels generated

from self-supervised stereo-matching methods. The approach involves estimating the confidence map of the pseudo-ground truth depth map to mitigate the degradation caused by inaccurate pseudo-depth maps. To address the prediction error of the confidence map, a threshold network is introduced to dynamically learn the threshold conditioned on the pseudo-depth maps. The filtered pseudo-depth labels are then used to supervise the monocular depth network. Additionally, a probabilistic refinement module based on the pixel-adaptive convolution (PAC) layer is proposed to refine the monocular depth map using its uncertainty map.

Peng et al. [129] addressed the need for cost-effective depth estimation methods and explored how the potential capacity of self-supervised approaches can be fully harnessed without increasing the computational burden. The authors proposed three key contributions: (1) a data augmentation technique called data grafting, which aimed to appropriately weaken the relationship between depth and vertical image position, (2) an exploratory self-distillation loss, utilizing selective post-processing to generate enhanced self-distillation labels, and (3) the full-scale network architecture, designed to enhance the depth estimation task's specialization and improve the model's representational power.

C. Semi-Supervised Methods

An alternative way to estimate depth is through a semi-supervised approach, when only a small number of labeled data is available and a large number of unlabeled data are present. The labeled data can be provided through depth sensors, like in [165], or artificially through synthetic data, like in [162], or external semantic proxy labels, like in [164]. However, additional information about depth is not always available in general. Table V summarizes the reviewed methods for semi-supervised methods in monocular depth estimation. The methods are sorted by the year of publication and denote with bold numbers the “winning” method in various measures and metrics³.

Kuznietsov et al. [165] proposed a novel learning model based on sparse ground-truth depth, while the deep network, which they proposed, produces photo-consistent dense depth maps. A future direction as suggested is to couple monocular and stereo-depth cues.

He et al. [163] presented a wearable-friendly system for monocular depth estimation from stereo images. The proposed loss function was the right-left consistency loss with the addition of sparse LiDAR data.

Ramirez et al. [164] demonstrated that integrating semantic information aids monocular depth estimation. They extended Godard et al.'s [142] architecture by adding a decoder for semantic segmentation. Their loss function incorporates depth, semantics, and cross-domain discontinuity terms, particularly enhancing depth predictions near object edges.

Zhao et al. [162] incorporated synthetic data into their monocular depth estimation method using image style translation and

³In these Tables, the terminology is as follows: *AbsRel*, *SqRel*, *RMSE*: Error metrics, as defined in Section II-D (lower is better); δ_1 , δ_2 , δ_3 : Accuracy metrics with thresholds, as defined in Section II-D (higher is better).

TABLE V
COMPARISON OF SEMI-SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS ON THE KITTI DATASET

Year	Method	Approach	AbsRel ↓	SqRel ↓	RMSE ↓	RMSE _{log} ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
2019	Zhao et al. [162] (K+vK)	Synthetic, style translation & stereo	0.149	1.003	4.995	0.227	0.824	0.941	0.973
2018	He et al. [163]	Weak-supervision and stereo	0.110	1.085	5.628	0.199	0.855	0.949	0.981
	Ramirez et al. [164]	Semantic labeling inclusion	0.143	2.161	6.526	0.222	0.850	0.939	0.972
2017	Kuznetsov et al. [165] (E)	Direct image alignment loss	0.113	0.741	4.621	0.189	0.862	0.960	0.986

depth estimation modules. Image style translation, based on CycleGAN [168], transforms between real and synthetic images. This integration of synthetic data supplements depth information not captured by unsupervised approaches.

IV. DISCUSSION AND CONCLUSION

Monocular depth estimation presents unique challenges compared to stereo methods, as demonstrated by Smolyanskiy et al. [169]. While monocular algorithms struggle due to the absence of stereo information, stereo approaches leverage multiple images for accurate depth recovery. Notwithstanding, the scientific community shows substantial fascination in monocular depth estimation, as evidenced by the numerous and diverse methods that have been suggested. This up-to-date review has assessed more than 80 state-of-the-art methods that encompass the progress achieved in this domain over the span of almost a decade. Depending on the learning approach, the methods were divided into three primary groups: supervised, self-supervised, and semi-supervised. The methods were described individually and their performance was presented in comparative tables. Early approaches, such as those dating back to 2014, predominantly relied on traditional CNN architectures. The paradigm shifted towards encoder-decoder architectures in the supervised domain, a transition that facilitated richer feature extraction and context modeling. Concurrently, self-supervised techniques embraced the integration of GANs as they achieved higher performance. Throughout the years, fusing multiple tasks, such as surface normal estimation with semantic labeling, has consistently worked well, although this often includes different types of data and integrating multiple networks. Nowadays, an emerging trend involves the application of vision transformers in supervised and self-supervised approaches.

A. Challenges and Limitations

Nevertheless, a key finding is that it is not yet clear how the relevant models estimate depth, as a result of the behavior of Deep Learning methods not being *explainable*, either in the classical terminology or in terms of the known theories on human depth estimation. By extension, the question of whether there is a relationship between the features recognized by neural networks and the corresponding features recognized by humans (depth cues) remains unanswered. While attention maps and gradient-based methods can provide insights into what parts of an image contribute to a particular prediction, the black-box nature of deep learning models can make it challenging to achieve human-level explainability that directly corresponds to the way humans perceive depth cues. Research on the above topics may

reveal weaknesses and limitations of current approaches, paving the way for new innovative methods with higher performance (accuracy, generalization, and real-time performance). At this point, it is worth mentioning the work of Dijk et al. [170], which examines how neural networks estimate depth from monocular images. Testing on four neural networks showed that the vertical position of the objects, and not the apparent size, plays an important role. Additionally, generalization appeared to be limited and dependent on very specific factors such as the objects' shadows. Similarly, Hu et al. [171] proposed a visualization method for CNNs about high-importance regions on images. Applying this method to state-of-the-art CNN architectures showed that the edges that define the geometry of the scene, the interior regions of individual objects, and the regions around vanishing points are essential for depth estimation. However, the above observations are far from an approach that will show what is, if any, the relationship between the features recognized by neural networks and visual depth cues. A direct link to this is the training data which has some, but not all, of the depth cues. Other approaches such as that of Tang et al. [172] focused exclusively on extracting information through defocusing. Another key challenge lies in the diversity and quality of data. An inherent limitation of data-driven approaches is their dependency on the training data, which might lack generalization to new and complex scenes. In addition, intrinsic limitations within monocular depth estimation methods come to the forefront, particularly in scenarios prone to optical illusions, reflective/translucent objects and abstract representations, introducing depth prediction ambiguities. These issues, while inherent to monocular depth estimation, emphasize the importance of *robustness and reliability in real-world applications*. These challenges underscore the need for *large, diverse, and well-annotated datasets* to ensure robust and accurate depth estimation across a wide range of real-world scenarios.

B. Future Directions

Vision transformers and attention mechanisms are highly relevant to addressing monocular depth estimation challenges [39]. Potentially, vision transformers' self-attention mechanism may capture intricate depth cues, such as texture gradients and occlusion patterns, often missed by traditional CNNs, but this has to be proven. Additionally, attention mechanisms enhance accuracy by focusing on informative regions, aiding in capturing long-range dependencies between objects. These techniques are potentially expected to offer better explainability through attention maps, shedding light on the neural network's depth perception process and its alignment with human visual cues. By incorporating these methods, monocular depth estimation

can achieve enhanced accuracy, explainability, and robustness across diverse scenes. Future research directions also involve refining loss functions, hyper-parameter learning, and intrinsic matrix estimation. In addition to transformers, graph convolution [173], knowledge distillation [174], 3D convolution [175] and attention mechanisms [176] hold promise. Geometry and semantics integration, such as multi-task training and geometric constraints, can enhance performance [132].

In conclusion, monocular depth estimation is a field with ongoing challenges and promising directions. This review aims to contribute to mapping the domain and identifying the path forward.

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